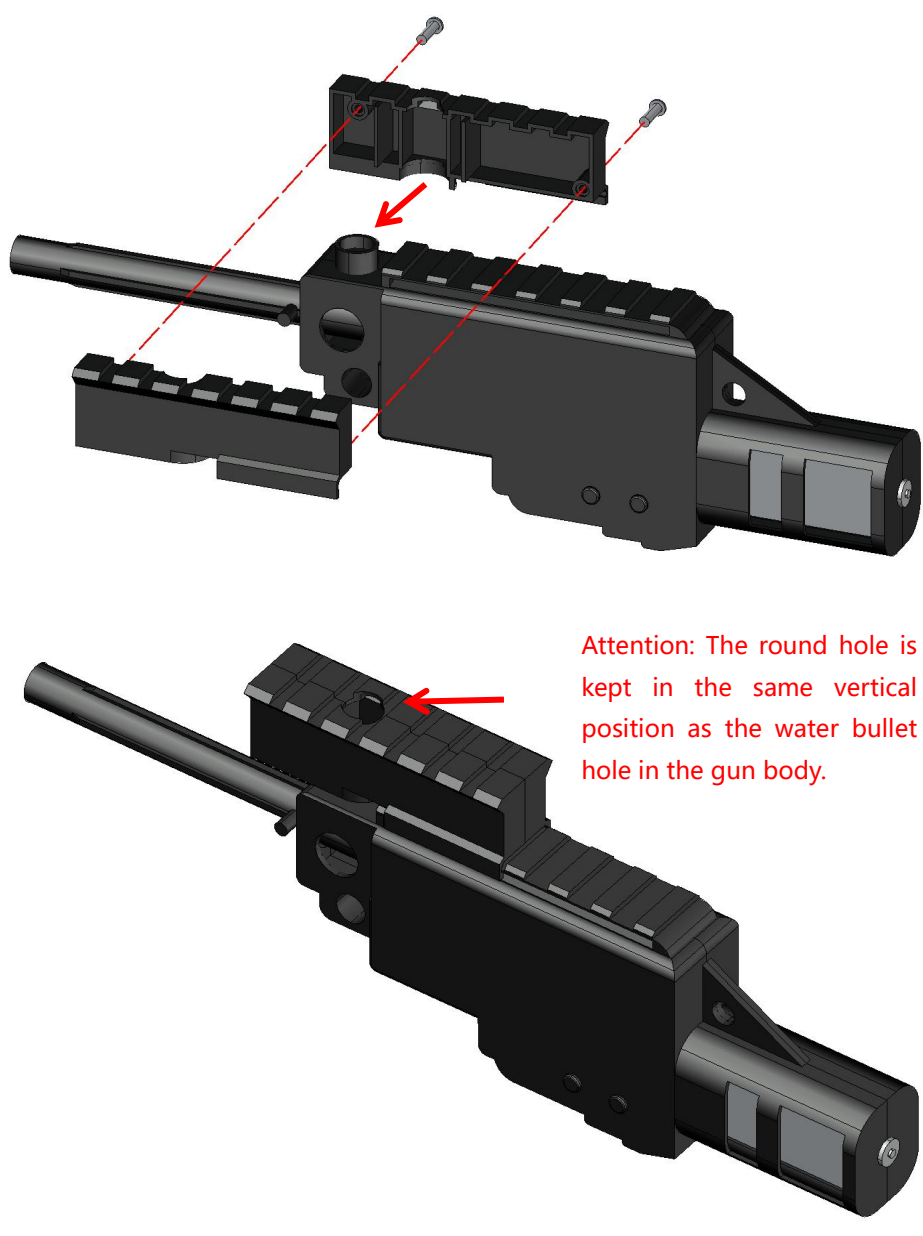


Step 1 Assemble the water bomb gun

Parts List	Black water bomb gun*1	Aim mirror base*2	M2.5*6mm Round head screw*2
Splicing Diagram	 Attention: The round hole is kept in the same vertical position as the water bullet hole in the gun body.		

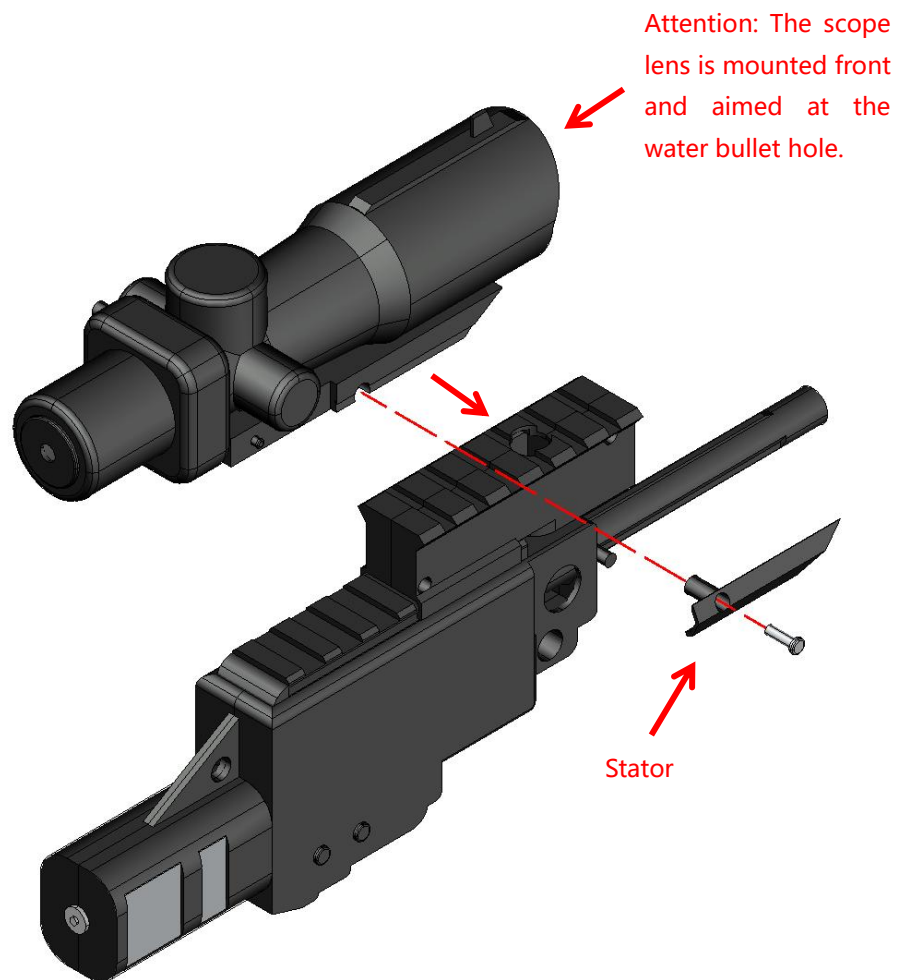
Step 2 Assemble the aim mirror

Parts List

Aim mirror*1

Stator *1

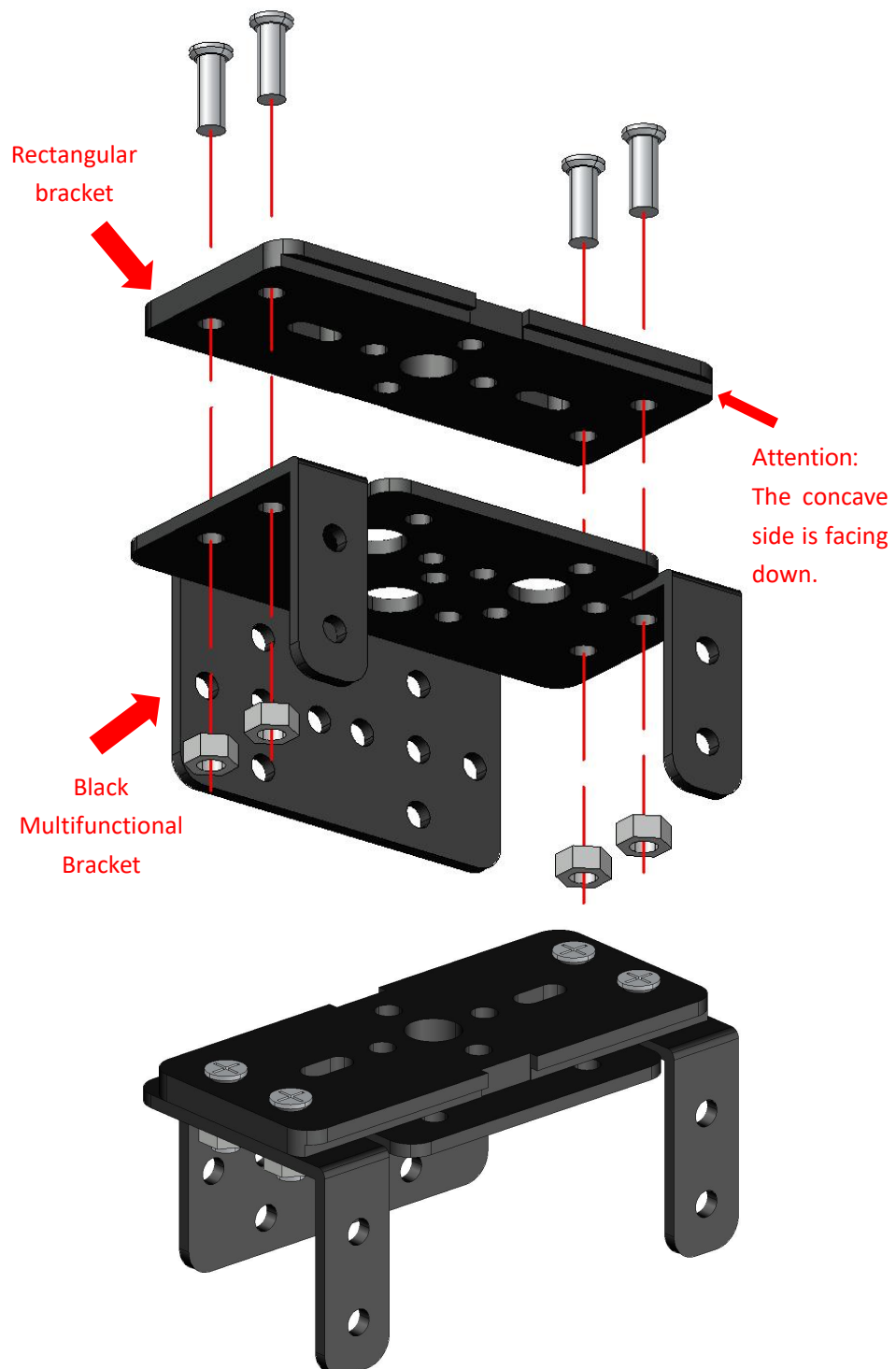
M2.5*6mm Round head screw*1

Splicing
Diagram

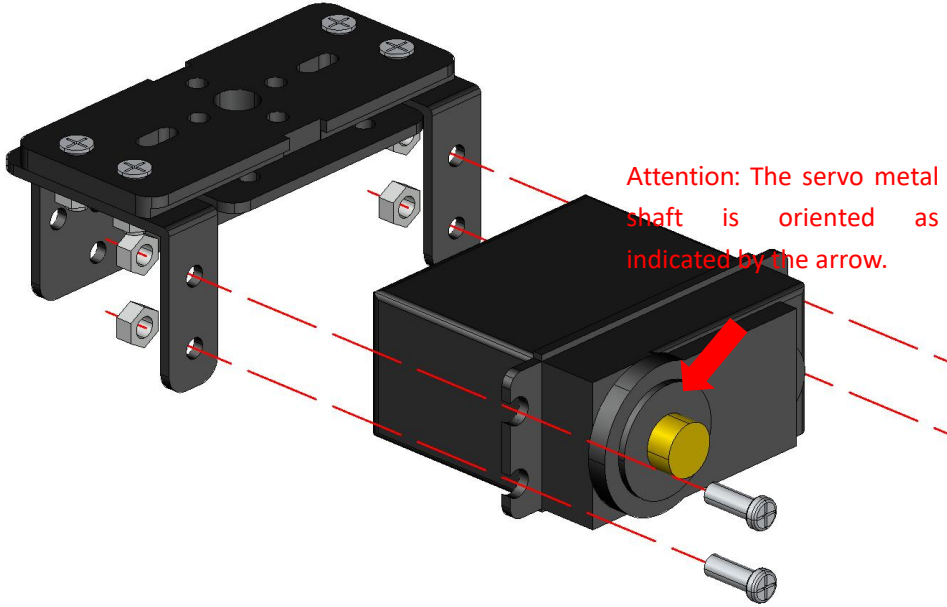
Step 3 Install fixed bracket

Parts List	Rectangular bracket*1	Black Multifunctional Bracket*1
	M3*10mm Flat Head screw*4	M3 nut*4

Splicing
Diagram



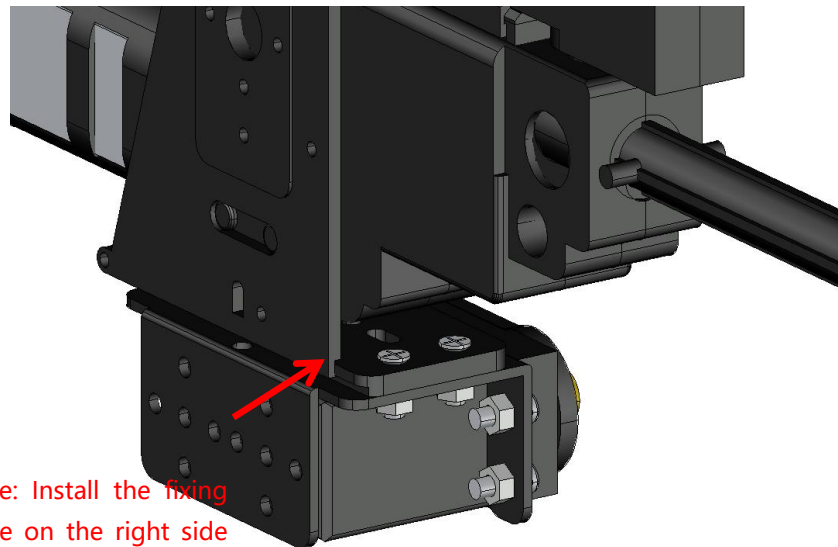
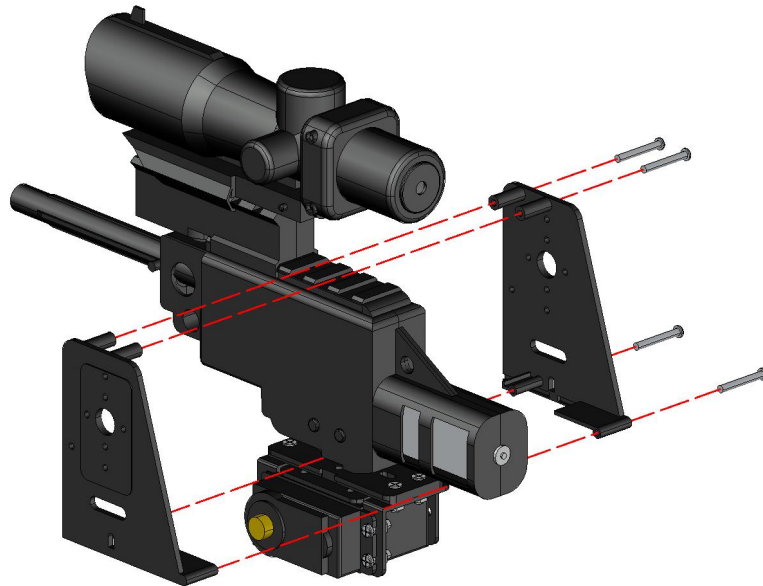
Step 4 Assemble the MG995 black servo module

Parts List	M3*10mm Round head screw*4	M3 nut*4	MG995 black servo module*1
Splicing Diagram	 Attention: The servo metal shaft is oriented as indicated by the arrow.		
Notes	MG995 black servo needs to have the status of the calibration program already uploaded. If it's not calibrated yet, go back to the must be read document of QD005 before use and recalibrate according to the prompts.		

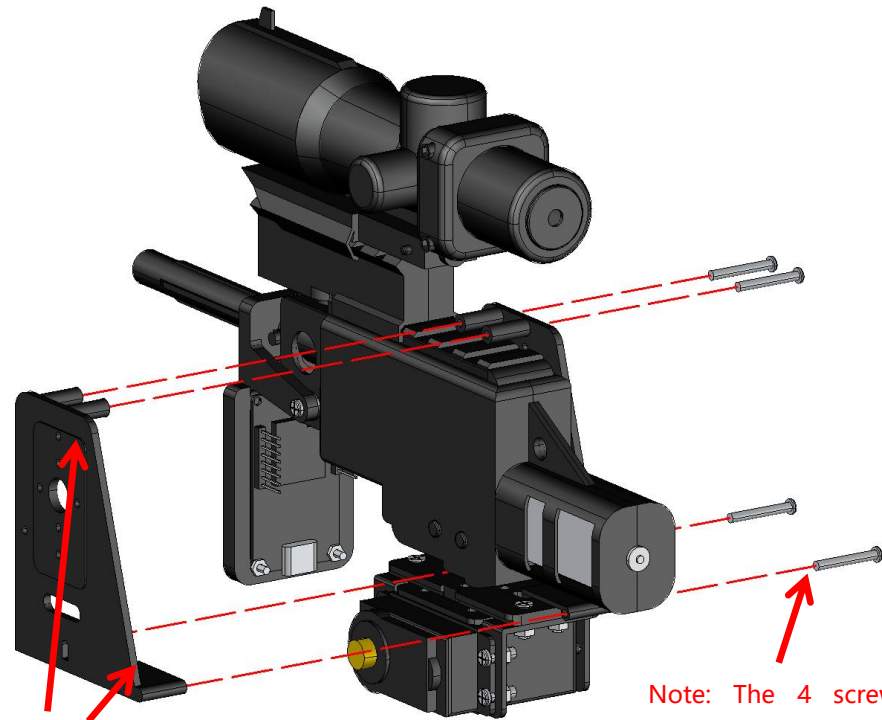
Step 5 Fixed water bomb gun

Parts	Water bomb gun side fixed	M2.5*22mmHalf teeth self-attack
List	plate*2	screw*4

Splicing
Diagram

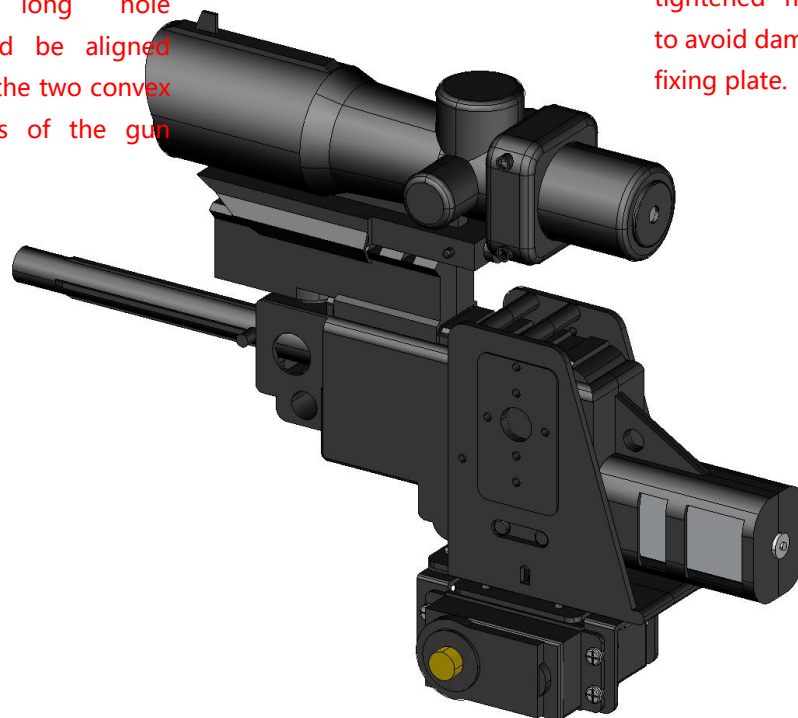


Note: Install the fixing plate on the right side of the gun body first, and then install the steering gear into the slot of the fixing plate.



Note: Install the left fixing plate, where the long hole should be aligned with the two convex points of the gun body.

Note: The 4 screws here do not need to be screwed too hard, but tightened moderately to avoid damaging the fixing plate.



Step 6 Assemble the servo steering plate

Parts List

Assemassembled water gun structure

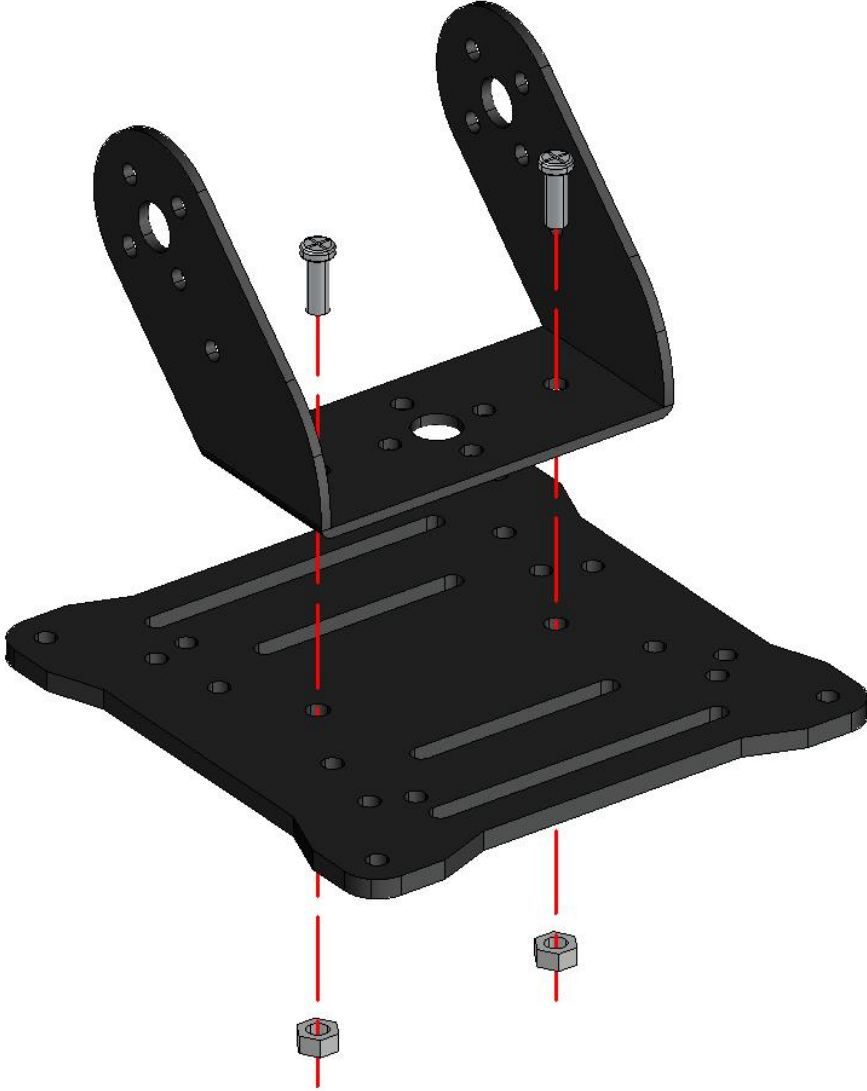
Round rudder plate*1

Splicing

Diagram



Step 7 Fixed the black oblique U holder

Parts List	Black oblique U holder*1	Expand the acrylic board*1	M3*10mm Round head screw*2
	M3 nut*2		
Splicing Diagram			
Notes	Tear off the acrylic film in advance		

Step 8 Fixed water gun structure

Parts List

Assembled
water gun structureAssembled
oblique U holderM2*8mm Large flat
round head self-tapping
screw*2

Splicing

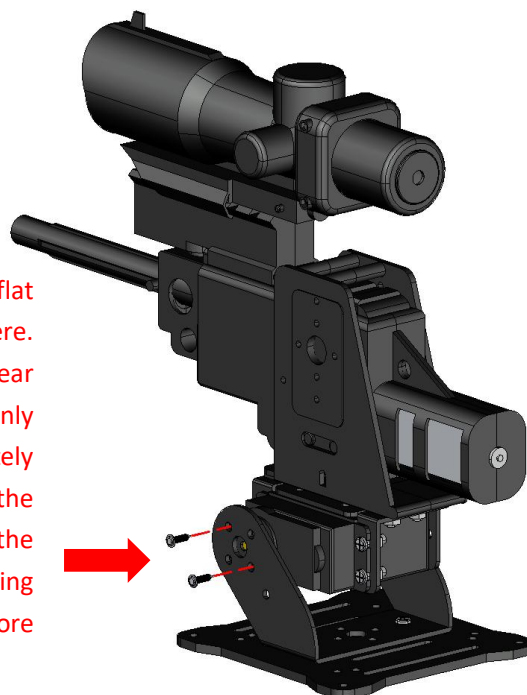
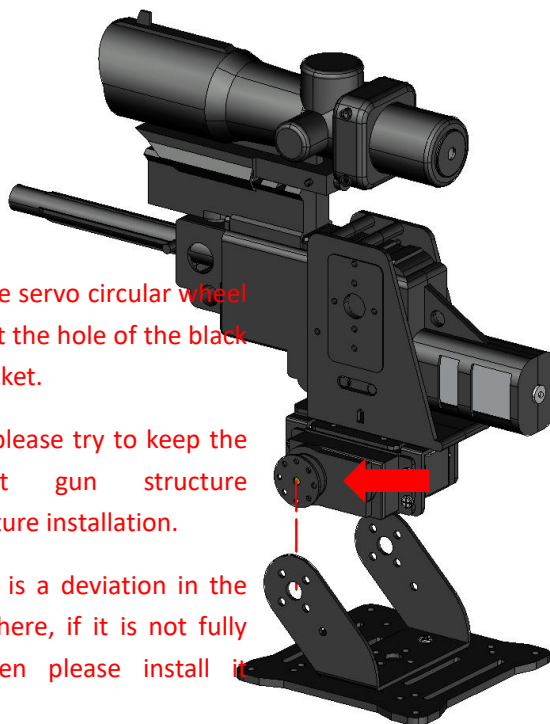
Diagram

Attention: The servo circular wheel is positioned at the hole of the black inclined U bracket.

At this point, please try to keep the water bullet gun structure horizontal posture installation.

Because there is a deviation in the hole position here, if it is not fully horizontal, then please install it above.

Attention : M2*8mm large flat head is screws diagonally here. Carefully check the steering gear holes during installation, only four holes can completely correspond to the holes of the bracket. If not, remove the steering wheel on the steering gear and reinstall it before operating.



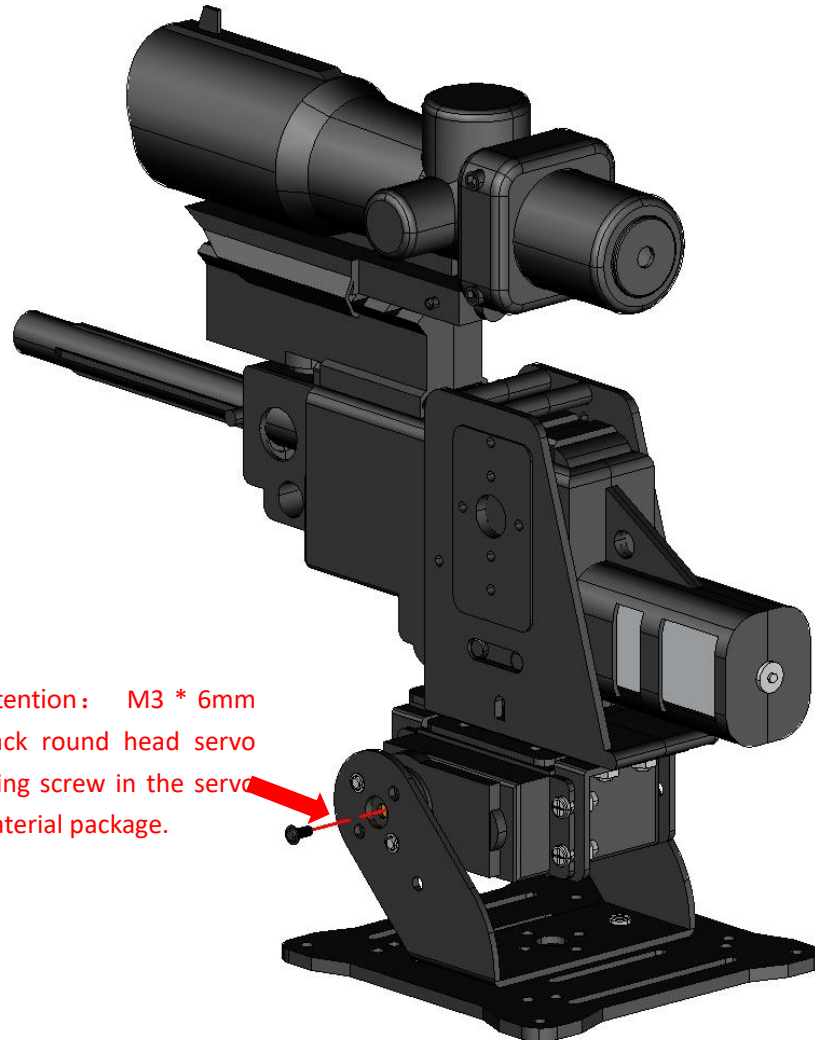
Step 9 Assemble the servo-fixing screw

Parts List

M3*6mm Black round-head servo-fixing screw*1

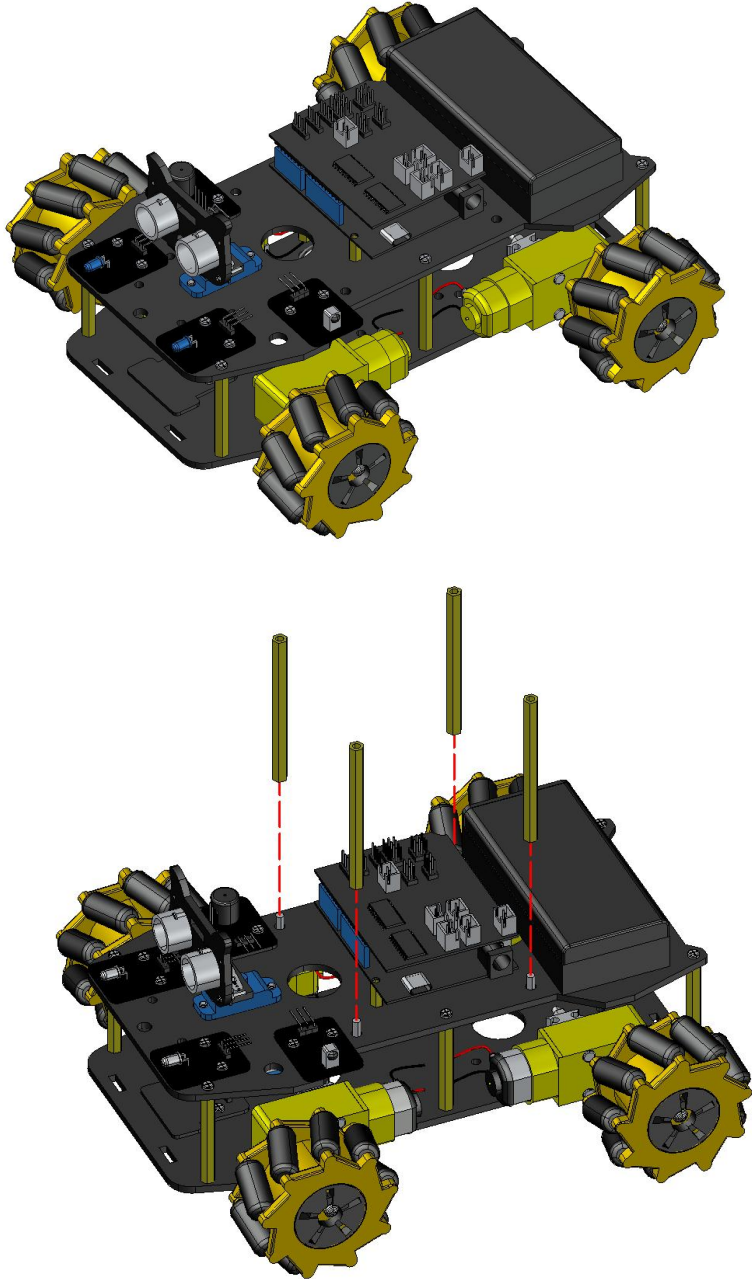
Splicing

Diagram

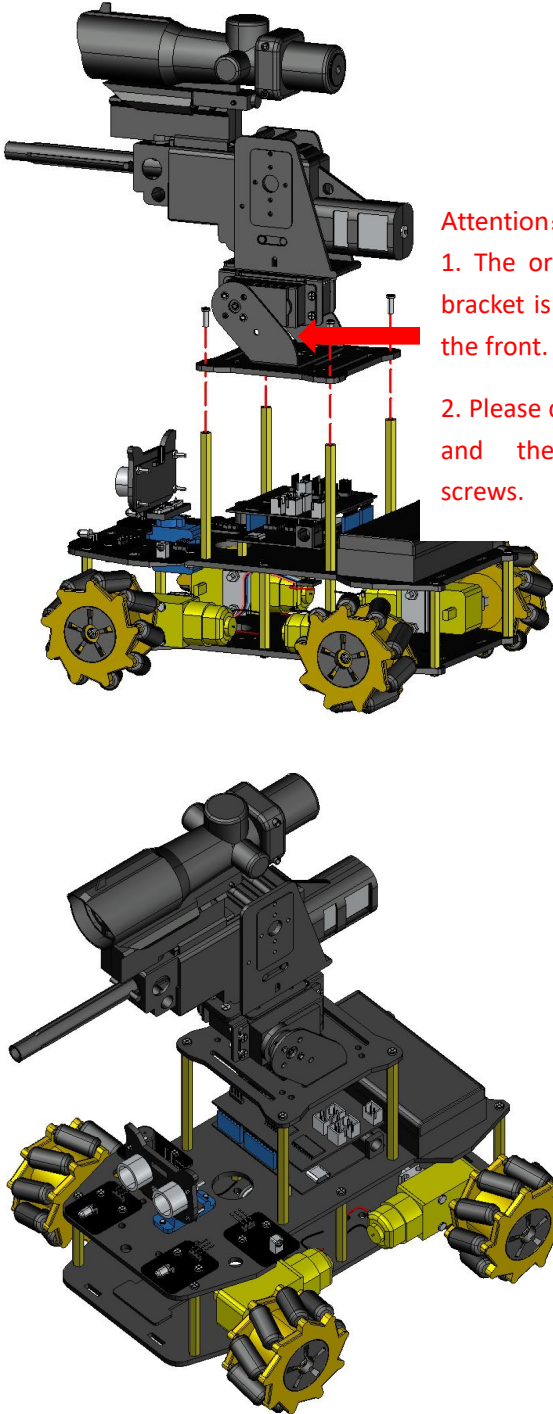


Attention: M3 * 6mm
black round head servo
fixing screw in the servo
material package.

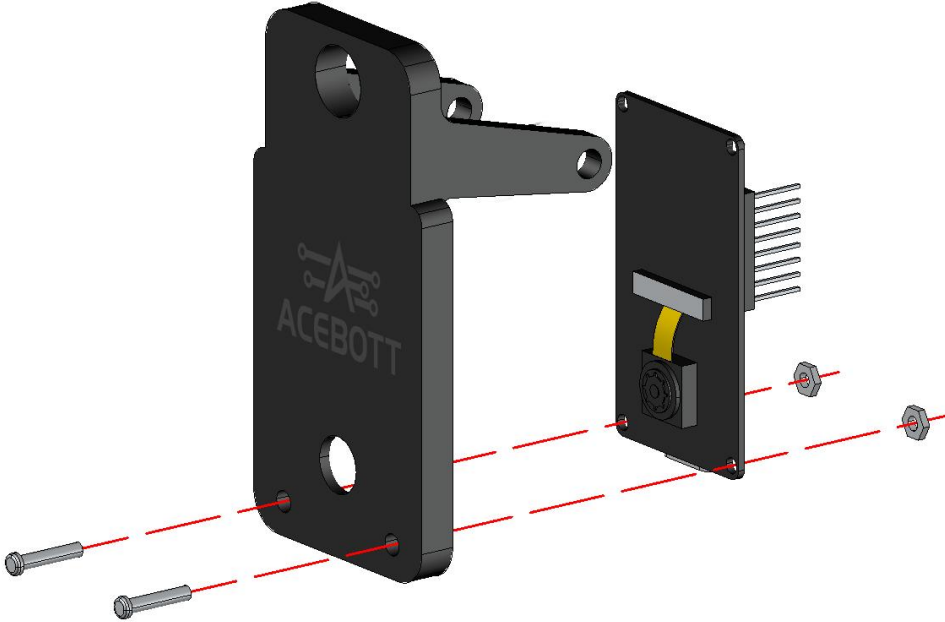
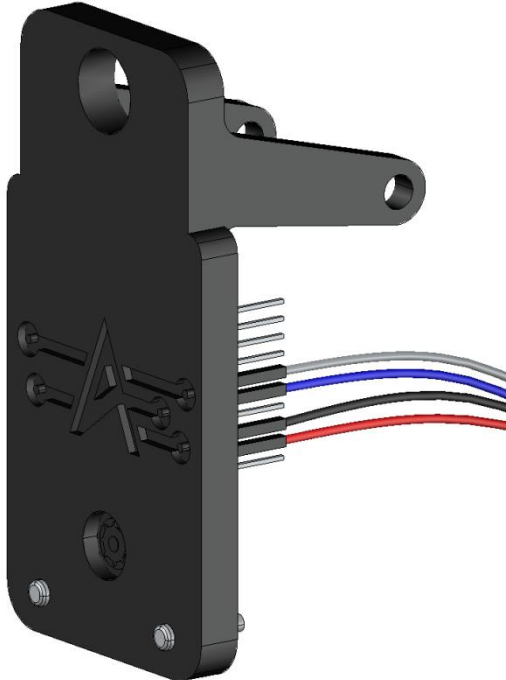
Step 10 Add copper columns

Parts List	Complete car	M3*65mm Double-pass copper column*4	M3*10mm Round head screw*4
Splicing Diagram			
Notes	Pass the screw from under the hole, hold it with the finger, and then tighten the M3 * 65mm double-pass copper column with the other hand.		

Step 11 Assemble the water bullet gun onto the car body

Parts List	M3*10mm Round head screw*4	Assembled water gun structure
Splicing Diagram	 Attention: 1. The orientation of the bracket is the direction of the front. 2. Please connect the wire and then install the screws.	
Notes	Please refer to the Step15 wiring diagram to complete the wiring of the water bullet gun trolley before installing the screws.	

Step 12 Assemble the camera module

Parts List	Camera module*1	Camera 3d printed holder*1
	M2*8mm Round head screw*2	M2 nut*2
Splicing Diagram		
		

Step 13 Fixed camera module

Parts List

M3*35mm Round head screw*1

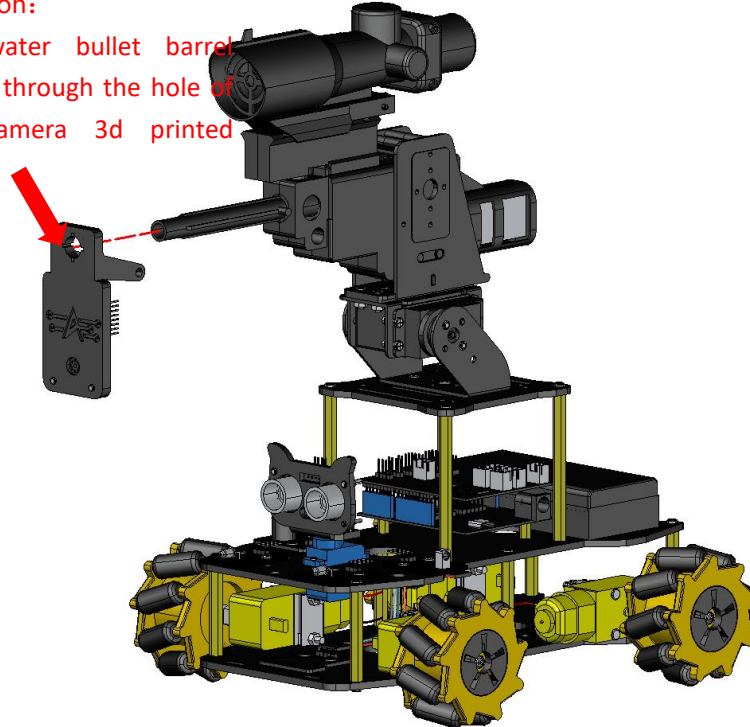
M3 nut*1

Splicing

Diagram

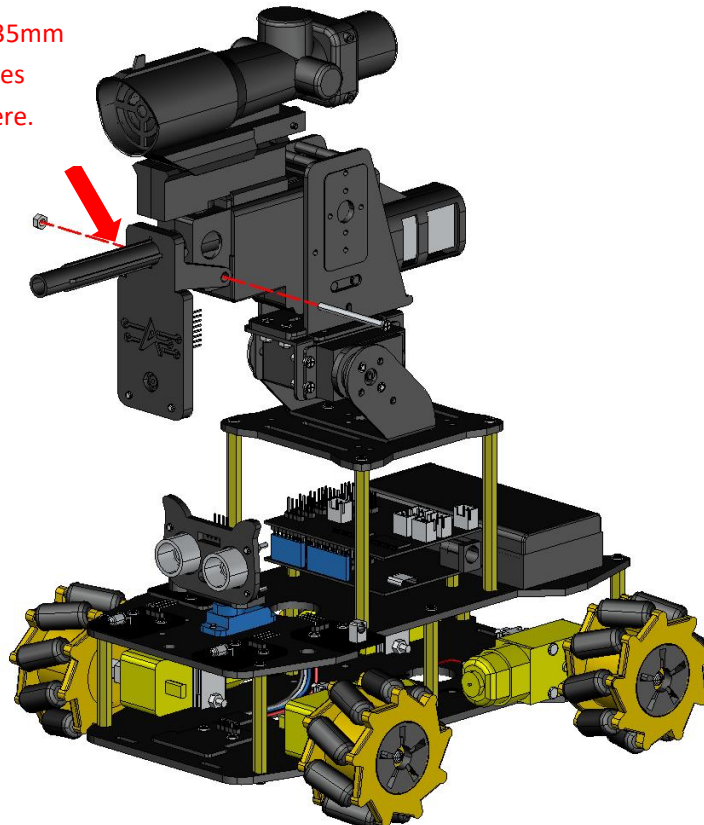
Attention:

The water bullet barrel passes through the hole of the camera 3d printed holder.



Attention:

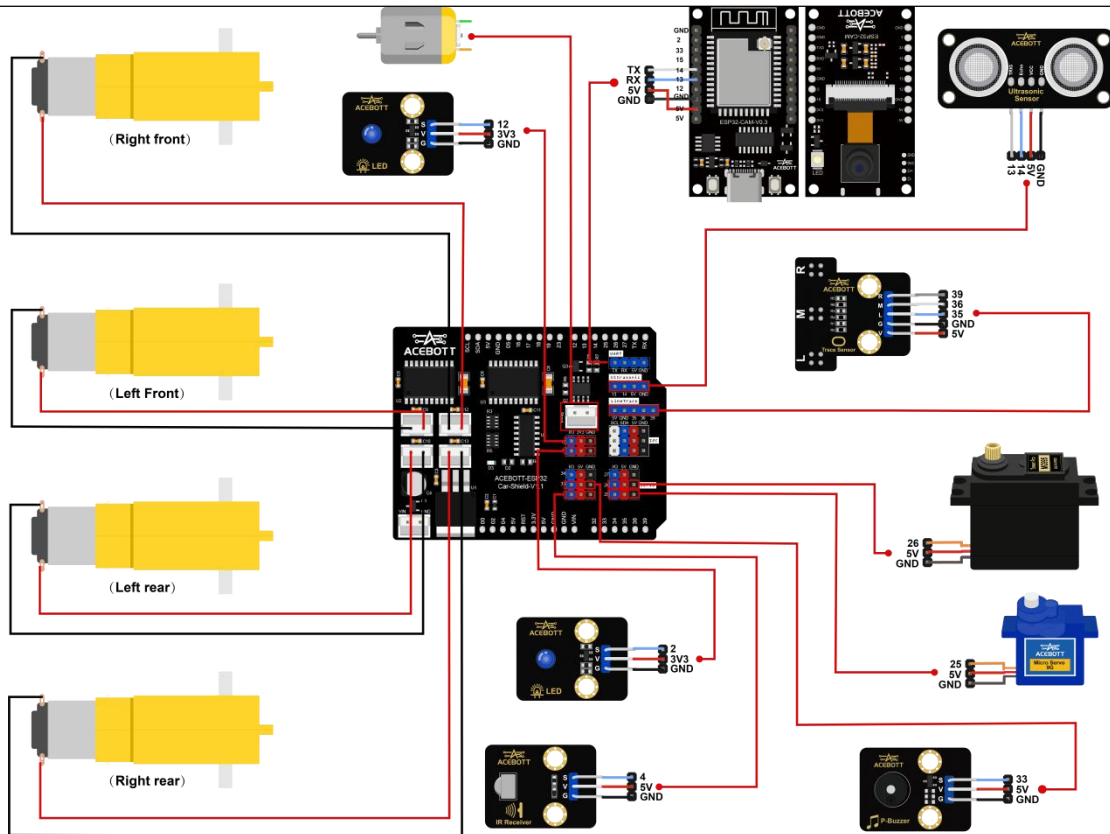
The M3 * 35mm screw passes through here.



Step 14 Assembly completed



Step 15 Wiring



1. The DuPont wires are blue, red, and black. The blue wire connects to the S pin, the red wire to the V pin, and the black wire to the G pin.

2. The servo wire colors differ from standard DuPont wires: the red wire connects to the V pin, the brown wire to the G pin, and the yellow wire to the S pin.

3. For the ultrasonic sensor wires: the black wire connects to the GND pin, the red wire to the V pin, the blue wire to the ECHO pin, and the white wire to the TRIG pin.

4. For the line-following sensor wires: the red wire connects to the 5V pin, the black wire to the GND pin, the blue wire to the L pin, the white wire to the M pin, and the gray wire to the R pin.

5. Pin 14 of the camera module is connected to the TX pin of the esp32 controller board, and pin 13 is connected to the RX pin of the esp32 controller board.

6. Be sure to connect the modules to the ESP32 control board strictly according to the wiring instructions. Incorrect wiring may cause a short circuit in the ESP32 control board and result in damage.